Land Use Classification of Satellite Imagery Using Transfer Learning with ResNet18

Introduction

Satellite imagery plays an important role in several real-world applications including environmental monitoring, agricultural management, urban planning, land-use mapping, and disaster response. Manual analysis of large volumes of satellite images is difficult and time-consuming, making automated classification approaches highly valuable.

This project addresses the problem of land-use classification using the EuroSAT RGB dataset. The task can be formulated as a supervised multiclass image classification problem in which the input consists of RGB satellite images and the output corresponds to one of ten land-use classes:

- AnnualCrop
- Forest
- HerbaceousVegetation
- Highway
- Industrial
- Pasture
- PermanentCrop
- Residential
- River
- SeaLake

Some classes present high visual similarity and therefore constitute a challenge for image classification systems. For example, HerbaceousVegetation, Pasture, and PermanentCrop frequently share similar textures and color distributions. Additionally, satellite imagery presents variability caused by illumination conditions, seasonal effects, and geographic differences, increasing intra-class variability.

Instead of training a convolutional neural network from scratch, a transfer learning approach using a pre-trained ResNet18 model was adopted to reduce computational requirements while adapting learned visual features to satellite imagery.

Data

The project uses the EuroSAT RGB dataset, a publicly available benchmark dataset for satellite image classification. The dataset contains Sentinel-2 RGB satellite images categorized into ten land-use classes.

Dataset characteristics

- Total number of images: 27,000
- Number of classes: 10
- Image type: RGB satellite imagery
- Original image dimensions: 64 x 64 pixels
- Task type: Multiclass classification

Data types

The EuroSAT dataset consists of unstructured image data. Unlike tabular datasets containing numerical and categorical variables, the predictors correspond to pixel intensity values distributed across three channels (red, green, and blue).

Data cleaning

No explicit data cleaning procedures such as missing-value imputation, duplicate removal, or outlier treatment were required because the dataset is already organized and labeled. No missing labels or corrupted images were identified.

Data preprocessing

Several preprocessing operations were applied before model training.

Resizing

Because pre-trained ResNet18 models expect larger inputs, images were resized from: 64 x 64 → 224 x 224 pixels.

Normalization

Image normalization was performed using ImageNet statistics:

- Mean: 0.485, 0.456, 0.406
- Standard deviation: 0.229, 0.224, 0.225

Normalization improves training stability and ensures compatibility with ImageNet pre-trained weights.

Data augmentation

Data augmentation techniques were applied only to the training dataset:

- Random horizontal flip
- Random rotation (15°)

These transformations increase data variability and reduce overfitting. Traditional feature engineering and feature selection were not required because CNNs automatically learn feature representations directly from image data.

Although the dataset is not perfectly balanced, class imbalance is relatively small and therefore no oversampling or class weighting methods were applied.

Data Organization

The original EuroSAT dataset is not initially divided into training, validation, and testing subsets. Therefore, the dataset was split into:

- Training set: 70%
- Validation set: 20%
- Test set: 10%

The training set is used for model learning, the validation set monitors performance during training and supports hyperparameter decisions, and the test set provides an unbiased estimate of model generalization.

Dataset organization

Dataset	Images
Training	18,900
Validation	5,400
Test	2,700

PyTorch Dataset and DataLoader objects were used to organize and batch the data. Training batches were shuffled to ensure that images from different classes were mixed during optimization, whereas validation and test batches were not shuffled because they are used only for evaluation.

A batch size of 32 images was selected.

One limitation of the adopted methodology is that data splitting was performed randomly. Neighboring satellite image patches may exhibit spatial correlation; therefore, geographic splitting could provide a stricter estimate of model generalization.

Methods

Transfer Learning with ResNet18

Transfer learning consists of adapting a model previously trained on a large dataset to a related task.

A pre-trained ResNet18 architecture was selected because it provides effective feature extraction while allowing adaptation to satellite imagery through fine-tuning.

Initially, all convolutional layers were frozen and only the final classification layer was trained.

The original ImageNet output layer was replaced with 10 output neurons corresponding to the EuroSAT classes.

Fine-Tuning Strategy

Training was performed in two phases.

Phase 1: Feature extraction

During the first phase, all convolutional layers remained frozen and only the final classification layer was trained. This phase allows the model to learn the relationship between extracted visual features and EuroSAT classes.

Phase 2: Fine-tuning

During the second phase, the final residual block (layer4) was unfrozen and a smaller learning rate was used.

This strategy enables adaptation of high-level visual representations to satellite imagery while preserving previously learned knowledge.

The optimizer was redefined so that both the classifier and the unfrozen residual block could update their parameters.

Hyperparameter Selection

Hyperparameters were selected empirically based on training stability and validation performance.

Hyperparameter	**Value**
Batch size	32
Optimizer	SGD
Momentum	0.9
Learning rate (Phase 1)	0.001
Learning rate (Phase 2)	0.0001
Epochs (Phase 1)	5
Epochs (Phase 2)	10
Loss function	CrossEntropyLoss
Image size	224 x 224

Cross-entropy loss was selected because it is commonly used for multiclass classification tasks.

Model checkpointing was implemented to save the weights associated with the highest validation accuracy.

Results

Training and validation performance were monitored throughout the learning process.

During Phase 1, validation accuracy reached approximately 88%. After fine-tuning the final residual block, performance improved further and validation accuracy increased to approximately 95%.

Final performance on the test dataset:

- Test accuracy: ~95.4%
- Macro F1-score: ~0.95

Forest, Residential and SeaLake achieved the strongest performance, whereas HerbaceousVegetation, PermanentCrop and Highway showed slightly lower scores due to visual similarity among classes.

Learning curves showed decreasing loss and increasing accuracy throughout training, while the confusion matrix exhibited strong diagonal values indicating that most images were correctly classified.

Manual prediction was additionally performed by selecting a random image from the test set and comparing the predicted class with the true label. The example prediction correctly classified the selected image.

Analysis

The obtained results demonstrate that transfer learning is highly effective for satellite image classification.

Fine-tuning further improved model performance, increasing validation accuracy from approximately 88% to approximately 95%, indicating that adapting higher-level features contributes to learning more task-specific representations.

Learning curves indicate stable convergence with a small gap between training and validation performance. Training accuracy approached approximately 94% while validation accuracy remained near 95%, indicating limited evidence of overfitting.

The confusion matrix revealed that most prediction errors occurred among vegetation-related classes, particularly HerbaceousVegetation, Pasture and PermanentCrop.

This behavior is expected because these categories frequently share similar texture patterns and color distributions.

Distinct classes such as Forest and SeaLake achieved stronger performance due to clearer visual patterns.

Limitations

Although strong results were achieved, several limitations should be acknowledged:

- the dataset split was performed randomly rather than geographically;
- only a single architecture (ResNet18) was evaluated;
- hyperparameters were selected empirically rather than through systematic optimization techniques.

Future work

Potential future improvements include:

- evaluating deeper architectures such as ResNet50 or EfficientNet;
- applying systematic hyperparameter optimization techniques;
- testing satellite-specific foundation models.

References

PyTorch Documentation.

EuroSAT RGB Dataset (Kaggle).

Contributions

Catarina Silva: Data preparation, model development, experimentation, and documentation
João Alexandre: Deployment implementation, experimentation, result analysis, and documentation